

Exploratory Data Analysis for Liquefaction Assessment

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- 2 Understanding Distributions
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The Foundation of Machine Learning

Before building models, understand your data!

EDA answers:

- What does the data tell us about the physics?
- Are there quality issues (missing, errors, outliers)?
- Which features predict lateral spreading?
- How should we preprocess?

Gilbert Strang's approach:

"See the structure first, then calculate."

Liquefaction and Lateral Spreading

Physical process:

- 1 Earthquake shaking
- 2 Pore pressure increases
- 3 Effective stress drops: $\sigma' = \sigma - u$
- 4 Soil loses strength
- 5 Lateral displacement occurs

Dataset: Durante & Rathje (2021)

- 7,291 case histories
- Target: $\geq 0.3\text{m}$ displacement = spreading

The Key Question

Can we distinguish spreading from non-spreading sites?

Approach: Compare distributions by class

- **Overlapping distributions:** Feature not useful
- **Separated distributions:** Feature is predictive

Best visualization: Overlaid histograms

- See both classes simultaneously
- Directly compare shapes and locations
- Immediate intuition about separability

This is more informative than looking at all data together!

Reading Overlaid Histograms

What to look for:

- ① **Separation:** How far apart are the peaks?
 - Far apart → strong predictor
 - Overlapping → weak predictor
- ② **Direction:** Which class has higher values?
 - GWD: No-spread sites have higher values (deeper water table = safer)
 - PGA: Spread sites have higher values (more shaking = more risk)
- ③ **Overlap area:** How much do they overlap?
 - Small overlap → easy to classify
 - Large overlap → difficult boundary

Why Quartiles Matter

Problem: Mean and std are sensitive to outliers

Example: Water depths (m): {1.5, 1.8, 2.0, 2.1, 2.3, 15.0}

- Mean = 4.12 m (misleading!)
- Median = 2.05 m (representative!)

Quartiles divide data into four equal parts:

- Q1 (25th percentile): 25% below
- Q2 (Median, 50th percentile): middle value
- Q3 (75th percentile): 75% below
- IQR = Q3 - Q1 (middle 50%)

Box plots visualize the five-number summary

Example: Ground Water Depth (GWD)

From our data:

- $Q1 = 1.64$ m, Median = 1.98 m, $Q3 = 2.46$ m
- $IQR = 0.82$ m

Engineering interpretation:

- 50% of sites: water table at 1.64-2.46 m depth
- Sites with $GWD \leq Q1$ (1.64 m): **High risk** zone
- Sites with $GWD > Q3$ (2.46 m): **Lower risk** zone

Action: Quartiles create natural risk categories!

In geotechnical engineering, outliers can be:

- ➊ **Measurement errors:** Remove
 - Negative water table depth
 - Physically impossible values
- ➋ **Extreme events:** Keep!
 - Major earthquakes
 - Unusual but valid site conditions

Never blindly remove outliers!

- They might be your most important data
- Use engineering judgment + statistics

IQR Method (Most Robust)

Rule: Values beyond these bounds are outliers

$$\text{Lower bound} = Q1 - 1.5 \times \text{IQR}$$

$$\text{Upper bound} = Q3 + 1.5 \times \text{IQR}$$

Why $1.5 \times \text{IQR}$?

- John Tukey's empirical rule
- For normal distribution: captures $\sim 99.3\%$ of data
- Balances sensitivity vs over-flagging

Advantages:

- Works for any distribution shape
- Not fooled by outliers themselves
- Easy to explain to engineers

What is a Z-Score?

Z-score = how many standard deviations from the mean

$$z = \frac{x - \mu}{\sigma}$$

Interpretation:

- $z = 0$: At the mean
- $z = 1$: One std above mean
- $z = -2$: Two std below mean
- $|z| > 3$: Outlier (only 0.3% of normal data)

68-95-99.7 Rule:

- 68% within $\pm 1\sigma$
- 95% within $\pm 2\sigma$
- 99.7% within $\pm 3\sigma$

Use 1: Outlier Detection

Example: Ground Water Depth

Data: Mean $\mu = 2.08$ m, Std $\sigma = 0.64$ m

Value	Z-score	Assessment
2.08 m	0	Average
2.72 m	1	Normal
3.36 m	2	Unusual
6.00 m	6.13	Outlier!

Threshold: $|z| > 3$ for outliers

Use 2: Standardization (More Important!)

Problem: Features have different scales

- GWD: 0.4 - 6.0 m
- PGA: 0.3 - 0.6 g

Solution: Standardize using z-scores

Transform each feature:

$$x_{\text{standardized}} = \frac{x - \mu}{\sigma}$$

Result: All features have mean = 0, std = 1

Why this matters for ML:

- Neural networks train better
- Distance-based methods (KNN, SVM) need equal scales
- Prevents large-value features from dominating

IQR vs Z-Score: When to Use Each

Method	Use For	Advantage
IQR	Outlier detection	Robust
Z-score	Outlier detection	Assumes normality
Z-score	Standardization	ML preprocessing

Recommendation:

- Detect outliers with IQR
- Standardize features with z-scores for neural networks
- Tree-based models don't need standardization

Measures linear relationship strength

Properties:

- Range: $r \in [-1, 1]$
- $r = 1$: Perfect positive correlation
- $r = -1$: Perfect negative correlation
- $r = 0$: No linear correlation

Interpretation for liquefaction:

- GWD vs Target: Negative (shallow water = more risk)
- PGA vs Target: Positive (more shaking = more risk)
- L vs Target: Negative (closer to river = more risk)

Warning: Correlation $\neq$ Causation!

Statistical Significance Testing

Question: Are distributions truly different, or just random chance?

Mann-Whitney U Test:

- Compares two distributions
- Non-parametric (no normality assumption)
- Returns p-value

P-value interpretation:

- $p < 0.001$: Extremely significant (***)
- $p < 0.01$: Very significant (**)
- $p < 0.05$: Significant (*)
- $p \geq 0.05$: Not significant

For our data: GWD, PGA, and L are highly significant ($p \leq 0.001$)

Before modeling, verify:

1 Missing values:

- How many? Which features?
- Delete ($\leq 5\%$) or impute ($\leq 5\%$)?

2 Physical plausibility:

- Negative values where impossible
- Values outside engineering bounds

3 Duplicates:

- Exact copies (data entry errors)?
- Same site, multiple events (valid)?

4 Class balance:

- 50-50 split? 90-10?
- Severe imbalance needs special handling

Must-know statistical tools:

1 Quartiles (Q1, Median, Q3):

- Robust alternative to mean/std
- IQR = middle 50% of data

2 Z-scores: $z = (x - \mu)/\sigma$

- Outlier detection: $|z| > 3$
- Standardization for ML

3 Correlation:

- Strength of linear relationship
- Not causation!

4 Statistical tests:

- Confirm differences aren't random
- P-value ≤ 0.05 = significant

What we learned:

1 Most important features:

- GWD: Strong negative correlation
- PGA: Strong positive correlation
- L: Moderate negative correlation

2 Data quality: Excellent

- No missing values
- All values physically plausible
- Minimal duplicates

3 Class balance: 60/40 split

- Reasonable balance
- Use stratified sampling

Ready for machine learning!

Based on EDA:

1 Preprocessing:

- Standardize (z-scores) for neural networks
- Raw features for tree-based models
- Stratified train-test split

2 Model selection:

- Start with Random Forest (robust, interpretable)
- Try XGBoost (often best performance)
- Neural networks with standardized inputs

3 Evaluation metrics:

- Multiple metrics: Accuracy, Precision, Recall, F1, AUC
- Confusion matrix
- Consider cost of errors in engineering context

"All models are wrong, but some are useful."

— George Box

"See the structure first, then calculate."

— Gilbert Strang

"Connect to the application - numbers must have physical meaning."

The goal is insight, not just numbers!

Common Mistakes to Avoid

- ❶ **Blindly trusting summary statistics**
 - Always visualize!
- ❷ **Removing outliers without investigation**
 - Might be your most important data
- ❸ **Confusing correlation with causation**
 - Ice cream sales correlate with drowning... both caused by summer!
- ❹ **Ignoring class imbalance**
 - 99% accuracy meaningless for 1% rare class
- ❺ **Skipping domain knowledge check**
 - Does this make engineering sense?

EDA is the foundation of successful ML

We accomplished:

- Understood data structure and quality
- Identified predictive features
- Compared distributions by class
- Detected and evaluated outliers
- Gained physical insight into liquefaction
- Made informed preprocessing decisions

"No amount of sophisticated modeling can compensate for poor understanding of your data."

Thank you!

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Remember:

"Understanding the data is understanding the physics."